Contents


```
%HW4
```

Q1

```
for t=-9:0.5:9
    if t>=0
        y=-3*t^2+5;
        fprintf('y(%g)=%g\n',t,y)
    else
        y=3*t^2+5;
        fprintf('y(%g)=%g\n',t,y)
    end
end
```

```
y(-9)=248
y(-8.5)=221.75
y(-8)=197
y(-7.5)=173.75
y(-7)=152
y(-6.5)=131.75
y(-6)=113
y(-5.5)=95.75
y(-5)=80
y(-4.5)=65.75
y(-4)=53
y(-3.5)=41.75
y(-3)=32
y(-2.5)=23.75
y(-2)=17
y(-1.5)=11.75
y(-1)=8
y(-0.5)=5.75
y(0)=5
y(0.5)=4.25
y(1)=2
y(1.5)=-1.75
y(2)=-7
y(2.5)=-13.75
y(3)=-22
y(3.5)=-31.75
y(4)=-43
y(4.5)=-55.75
y(5)=-70
y(5.5)=-85.75
y(6)=-103
```

```
y(6.5)=-121.75
y(7)=-142
y(7.5)=-163.75
y(8)=-187
y(8.5)=-211.75
y(9)=-238
```

Q2

```
n=input('input which term you pick: ');
k=n;
x=1;
y=2;
if n<=2
    disp('n has to be greater than 2')
elseif mod(n,1)==0
    while n>2
    y=x+y;
    x=y-x;
    n=n-1;
    end
    if mod(k,100)==11
        fprintf('the %dth term of fibonacci sequence is %d\n',k,y)
    elseif mod(k,100)==12
        fprintf('the %dth term of fibonacci sequence is %d\n',k,y)
    elseif mod(k,100)==13
        fprintf('the %dth term of fibonacci sequence is %d\n',k,y)
    elseif mod(k,10)==1
        fprintf('the %dst term of fibonacci sequence is %d\n',k,y)
    elseif mod(k,10)==2
        fprintf('the %dnd term of fibonacci sequence is %d\n',k,y)
    elseif mod(k,10)==3
        fprintf('the %drd term of fibonacci sequence is %d\n',k,y)
    else
        fprintf('the %dth term of fibonacci sequence is %d\n',k,y)
    end
else
    disp('invalid number')
end
%input which term you pick: 3.3
%invalid number
%input which term you pick: 1
%n has to be greater than 2
%input which term you pick: 22
%the 22nd term of fibonacci sequence is 28657
%input which term you pick: 34
%the 34th term of fibonacci sequence is 9227465
```

```
Error using input
Cannot call INPUT from EVALC.

Error in HW4 (line 18)
n=input('input which term you pick: ');
```

Q3

```
n=input('enter values in vector form: ');
sum=0;
for i=1:length(n)
    sum=sum+(n(i)).^2;
end
rms=sqrt(1/length(n)*sum);
fprintf('the root-mean-square is %f\n',rms)
%enter values in matrix form: [10 5 2 5]
%the root-mean-square is 6.204837
```

Q4

```
n=input('enter values in vector form: ');
sum=0;
for i=1:length(n)
    sum=sum+1/n(i);
end
h=length(n)/sum;
fprintf('harmonic mean is %f\n', h)
% enter values in matrix form: [10 5 2 5]
% harmonic mean is 4.000000
```

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